TT-P-1181a
May 14, 1968
SUPERSEDING
Int. Fed. Spec. TT-P-001181 (COM-NBS)
September 9, 1966 and
Fed. Spec. TT-P-97b
April 16, 1965

FEDERAL SPECIFICATION

PAINT, STYRENE-ACRYLATE SOLVENT TYPE, TINTS AND DEEP TONES (FOR EXTERIOR MASONRY)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers styrene-acrylate paints for application to exterior masonry surfaces.

1.2 Classification. Paint shall be of the following types, as specified:

Type I - Tinted masonry paint (see table III for suggested formulation).
Type II - Deep tone masonry paint (see tables IV and V for suggested formulation).

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

TT-P-143 - Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking Of.
TT-P-442 - Pigment, Titanium Dioxide, (For Protective Coatings).
SS-S-346 - Siding, (Shingles, clapboard, and sheets), Asbestos-Cement.

Federal Standards:

Fed. Std. No. 141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.
Fed. Std. No. 595 - Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

FSC 8010

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:18Z
Check the source to verify that this is the current version before use.

TABLE I. Quantitative requirements

Characteristics	Requirements Types I and II	
	Min.	Max.
Pigment (percent by wt. of paint)	46	51
Vehicle (percent by wt. of paint)	49	54
Nonvolatile vehicle (percent by wt. of vehicle)	30	35
Consistency K.U. at 77°F.	70	82
Absorption (inch)	--	1/4
Drying time:		
Set to touch (hr.)	1/4	3/4
Dry hard (hr.)	--	2
85 - Degree specular gloss (Sheen)	--	5
Fineness of grind	3	--
Weight per gallon (lbs.)	11.1	11.8
Styrene-acrylate resin (percent by wt. of nonvolatile vehicle)	50	--

3. REQUIREMENTS

3.1 Material. Materials shall be as specified herein. Material not definitely specified shall be of a good commercial quality entirely suitable for the purpose intended under normal conditions of use. Tinting pigments shall be light-fast and alkali resistant. The paint shall be free from material which will be toxic to personnel under normal conditions of use.

3.2 Composition. The composition of the paint shall be as specified herein.

3.2.1 The quantitative requirements of the paint shall be as specified in table I. The pigment for type I paint shall contain a minimum of 20.0 percent titanium dioxide (TT-P-442, Type III). This is equivalent to 18.6 percent TiO_2. The remainder shall be a mixture of calcium carbonate, diatomaceous silica, talc, and mica with the required tinting pigments (see table III). No quantitative requirements are specified for pigment composition for type II paints. However, suggested formulations for some colors are given in tables IV and V.

3.3 Qualitative requirements.

3.3.1 Condition in container. The paint as received shall be well ground and not caked, livered, thickened, skinned or badly settled in the container. Any settling shall be capable of being easily broken up and redispersed in the liquid portion to form a smooth paint of good brushing consistency when tested as specified in 4.4.1.

3.3.2 Color. When tested as in 4.4.7 the color shall match the reference standard such as in Fed. Std. No. 595 or color specified by the procuring agency (see 6.2).

3.3.3 Brushing, flowing, and leveling. The paint shall show easy brushing, good flowing, and good leveling properties. The paint shall not pull, sag, or have a quick set under the brush when tested as specified in 4.4.1 at a spreading rate of 400 square feet per gallon.

3.3.4 Compatibility with thinner. There shall be no evidence of incompatibility of any of the ingredients of the paint when tested as specified in 4.4.2.

3.3.5 Self-lifting. The surface shall produce no lifting or other film irregularities after recoating when tested as specified in 4.4.1 and 4.4.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:18Z
Check the source to verify that this is the current version before use.

3.3.6 Flexibility. The paint shall show no cracking when subjected to the flexibility test specified in 4.4.1 and 4.4.8.

3.3.7 Skinning. The paint shall not skin within 48 hours in a 3/4 full container when tested as specified in 4.4.1.

3.3.8 Odor. The odor of the paint immediately upon opening the container and during application shall not be pungent, irritating, or disagreeable when tested as specified in 4.4.3.

3.3.9 Accelerated weathering resistance. When the paint is tested as specified in 4.4.1 and 4.4.9, the apint film shall show no more than slight chalking and no other film defect. Color change shall not exceed 2 units and the chalk rating shall be not less than 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract to order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.2 Sampling and inspection. Sampling and inspection shall be performed in accordance with method 1031 of Fed. Test Method Std. No. 141.

4.3 Inspection of preparation for delivery. The packaging, packing, and marking shall be examined to determine compliance with section 5 of this specification.

4.4 Test procedures.

4.4.1 The following tests shall be conducted in accordance with Fed. Test Method Std. No. 141, except as otherwise specified herein. The test methods are shown in table II.

TABLE II.

Characteristics	Requirement reference	Applicable test method	
		Fed. Test Method Std. No. 141	Paragraph reference
Condition in container	3.3.1	3011	---
Consistency	Table I	4281	---
Brushing, flowing and leveling	3.3.3	4321	---
Compatibility with thinner	3.3.4	----	4.4.2
Pigment content	Table I	4022	---
Vehicle (total vehicle in paint)	Table I	4052	---
Nonvolatile (in vehicle)	Table I	4041	---
Absorption	Table I	4421	---
Drying time	Table I	4061	---
85 - Degree specular gloss (sheen)	Table I	6103	---
Fineness of grind	Table I	4411	---
Color	3.3.2	4250	4.4.7
Self-lifting	3.3.5	6252	4.4.4
Flexibility	3.3.6	6221	4.4.8
Skinning	3.3.7	3021	---
Odor	3.3.8	----	4.4.3
Accelerated weathering resistance	3.3.9	6152	4.4.9
Weight per gallon	Table I	4184	---
Pigments:			
Titanium dioxide	3.2.1	7081	---
Vehicle:			
Styrene-acrylate determination	Table I	----	4.4.5
Resin identification	Table I	----	4.4.6

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:18Z
Check the source to verify that this is the current version before use.

4.4.2 Compatibility with thinner.* Pour 10 milliliters (ml.) of paint into a 100-ml. graduated cylinder and gradually stir in a thinner composed of equal parts by volume of an aromatic and aliphatic blend of solvents having a Kauri-Butanol value of 60 or higher, (the aromatic portion must be of a lower volatility than the aliphatic portion). Continue adding and stirring until the 100-ml. graduation is reached. Settling of pigment is disregarded in determining homogeneity.

4.4.3 Odor. Observe the odor immediately upon opening the container and during application for compliance in 3.3.8.

4.4.4 Self-lifting. Brush a coat (method 2141) of the material on a glass panel (method 2021) of approximately 1 mil dry film thickness and allow to dry for 24 hours at room temperature. Recoat the panel using the same method and after the film has dried for 24 hours examine for compliance as specified in 3.3.5.

4.4.5 Determination of styrene-acrylate resin

Apparatus: Supercentrifuge
Centrifuge
Vacuum oven
Double beam grating infrared recording spectrophotometer (spectral resolution 3 cm -1 or better over entire range at slow scan)
Rock salt plates for use in spectrophotometer

4.4.5.1 Determination of total extractables (styrene-acrylate resin plus chlorinated paraffins 40 percent and 70 percent [1:1 ratio]).

Weight 50 to 55g of well-mixed sample to the nearest mg using a weighing bottle and wash with C.P. benzene into a supercentrifuge bowl of 300 ml. capacity. Mix well and dilute with benzene to approximately 80 percent of capacity of the bowl. Benzene may be easily lost during centrifugation. Make sure the gasket is in good condition and the cover is sealed tightly. Centrifuge for 1/2 hour at 50,000 r.p.m. Cool the bowl externally before opening. Decant the liquid into a tared 250 ml. beaker containing three Berl saddles (as boiling stones) and evaporate to dryness on a steam bath. Extract the material remaining in the bowl by thoroughly mixing with benzene and proceeding as before. Decant into a second tared 250 ml. beaker containing three Berl saddles and evaporate to dryness. Generally the first benzene extract is free of pigment and the second extract has a small amount. The presence of pigment in the decantate is indicated by a cloudy benzene solution.

When pigment is present in either extract, remove it by the following procedure. Evaporate the benzene extract to dryness on the steam bath, take up in benzene and transfer to two 100 ml. conical-shaped centrifuge tubes. Centrifuge at 2000 r.p.m. for 45 minutes and decant into the original beaker and again evaporate on the steam bath. If both extracts contain pigment, centrifuge in order, using the same centrifuge tubes to eliminate mechanical loss.

Dry beakers in a regular oven at 70 to 75°C over night. Then dry in a vacuum oven at 60 to 65°C and 30" of mercury (Hg) to constant weight (approximately 32 hrs.). Weigh beakers and calculate total weight of extractables.

When the total weight of the extracts is obtained, dissolve both in benzene and combine. Keep the benzene solution as concentrated as possible. From this benzene solution, make films on sodium chloride plates using a medicine dropper. Air dry films to remove excess benzene. Select 6 to 7 films of a thickness such that the absorbance maximum of the bands at 3.5 microns and 5.8 microns lies between 0.5 and 0.7. Vacuum dry the films over night or 16 hours at 60 to 65°C. and 30" of Hg. Scan the spectrum of each film from 2.5 to 15 microns using air as reference and a slow scanning speed. Draw the baselines as illustrated in Figures 1 and 2. Calculate the 5.8 μ / 3.5 μ ratio and from an average of 6 to 7 ratios, obtain the percent styrene-acrylate resin from a standard curve.

---

* Styrene-acrylate is not soluble in highly polar solvents such as acetone and alcohol.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:18Z
Check the source to verify that this is the current version before use.

free of wrinkles and raw edges. Varnished surfaces shall be ... be clean ar

4.4.5.2 Preparation of standard curve

The standard curve relates the 5.8 μ / 3.5 μ absorbance ratios to the percent styrene-acrylate resin. Prepare mixtures of styrene-acrylate resin (such as Pliolite AC) and chlorinated paraffins (such as Chlorowax 40 and Chlorowax 70) with the percent of total weight for styrene-acrylate at 10.00, 25.00, 50.00, 75.00, and 100. Since it is difficult to weigh chlorinated paraffin 40 percent to the nearest 0.1 mg, it is best to weigh this first in a weighing bottle, taking an arbitrary weight. Rinse into a 150 ml. beaker with benzene and add an exactly equal amount of chlorinated paraffin 70 percent. When this is dissolved, add a proportionate amount of styrene-acrylate resin to yield the desired percentage. Make certain that all materials are completely dissolved in benzene. Produce a series of films for each formulation as described in the preceding paragraph but with the absorbance maximum of the strongest band between 0.5 and 0.7. Dry them in the same manner and calculate the 5.8 μ / 3.50 μ ratios from their infrared spectrograms. Plot the average of the absorbance ratios for each formulation vs the percent styrene-acrylate resin using large graph paper. The plot is a curve and not a straight line.

4.4.5.3 Calculation: Total wt. of extractables x percent styrene-acrylate resin = total wt. of styrene-acrylate resin

$$\frac{\text{total wt. of styrene-acrylate resin} \times 100}{\text{wt. of paint sample}} = \text{percent styrene-acrylate resin in paint sample}$$

4.4.6 Identification of resin from vehicle(1)

Compare infrared spectrogram of the sample with the spectrogram of the known 50 percent styrene-acrylate resin mixture used in the preparation of the standard curve (4.4.5.2). See figure 2. There should be a close resemblance.

4.4.7 Color. A 3 mil wet drawdown film shall be made over a sealed and unsealed Morest Chart (or equal). After the film is dry only slight color difference shall be observed between the sealed and unsealed areas. In addition there shall be no streaking, spotting, or other irregularities. Determine the color in accordance with method 4250 of Fed. Test Method Std. No. 141 for compliance with 3.3.2.

4.4.8 Flexibility. Apply a film of paint with a Bird film applicator or equal to a wet film thickness of 0.002 inch to a tin panel (method 2012). Air dry for 18 hours, bake for 3 hours at 105°C $\pm$ 2°C. Cool for 1/2 hour and bend over a 1/8 inch mandrel. Examine for compliance as specified in 3.3.6.

4.4.9 Accelerated weathering resistance. Apply the paint by brushing at a rate of 400 square feet per gallon to the smooth side and edges of two asbestos-cement panels conforming to SS-S-346. Apply 2 coats allowing 24 hours drying between coats and 72 hours before testing. The first coat shall be reduced 8 to 1 parts by volume or 1 pint of thinner(2) to 1 gallon of paint. The second coat shall be applied without reduction. Determine the values of apparent reflectance with the amber, green, and blue filters before and after 300 hours of accelerated weathering. Examine the panels for chalking and other film defects, determine the color change by method 6123, and the chalk rating by method 6411. Compare the results for compliance with 3.3.9.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. The paint shall be packaged, packed, and marked in accordance with TT-P-143. The level of packaging shall be A, B or C and the level of packing shall be A, B, or C as specified (see 6.2). The paint shall be in 1-gallon metal cans, or 5-gallon steel pails as specified. (see 6.2).

6. NOTES

---

(1) Isolated vehicle see method 4032.

(2) See 4.4.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:18Z
Check the source to verify that this is the current version before use.

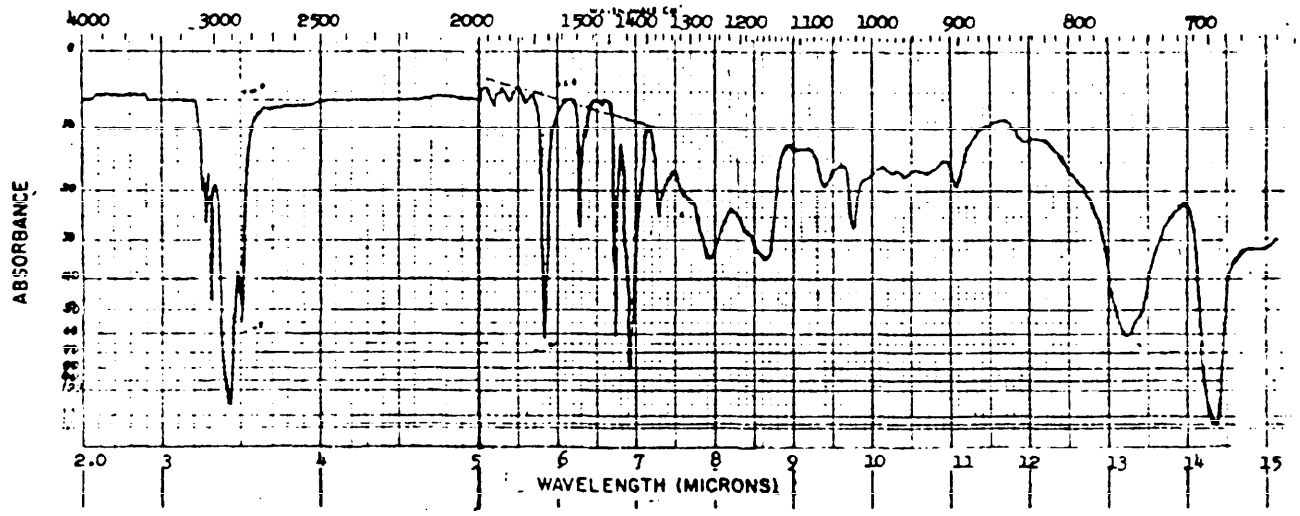


Figure 1 Extract of actual paint sample based on formulation in Table V.

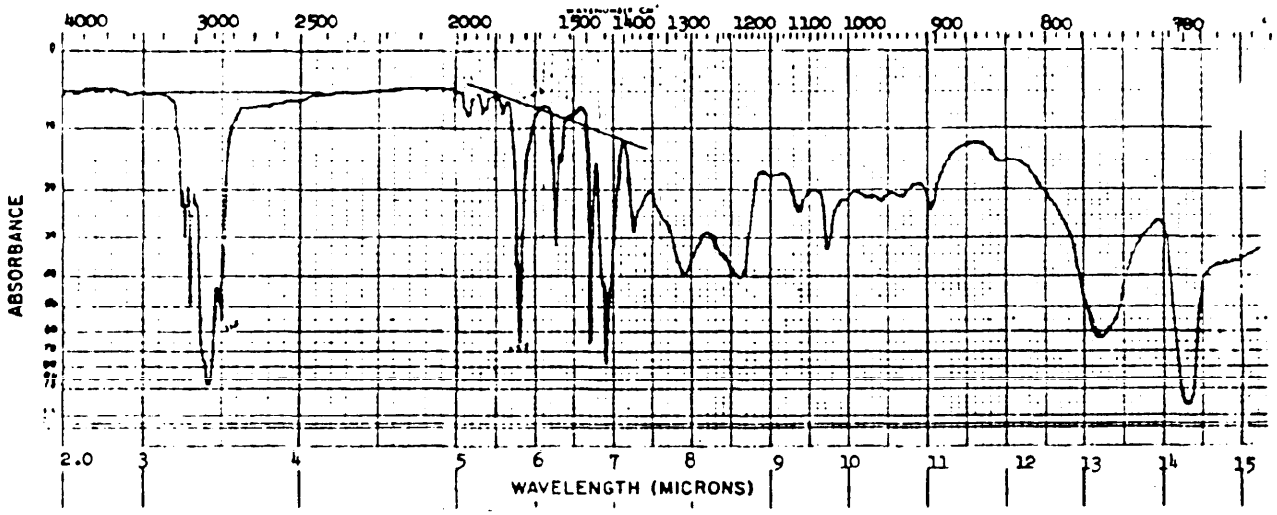


Figure 2. Mixture of styrene-acrylate resin, chlorinated paraffin 40% and chlorinated paraffin 70% in the proportion 50.00: 25.00: 25.00.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:18Z
Check the source to verify that this is the current version before use.

6.1 Intended use. This specification covers styrene-acrylate type I and type II tinted and deep tone paints previously covered by TT-P-97b types I and II for application to exterior concrete, stucco, and masonry surfaces which have been properly prepared. Surface preparation on work, no unusual preparation is required except in the case of new concrete, cinder block, or other extremely porous surfaces which should be filled or grouted. If filling or grouting is necessary, a suggested formulation in table VI is recommended. On previously painted surfaces, all loose powdery or flaking material, heavy chalking, and dirt should be removed by sand blasting, wire brushing, or scraping. It is further recommended that a two-coat system be used with the first coat, reduced 8 to 1 part by volume or 1 pint thinner (see 4.4.2) to 1 gallon of paint, and the second coat applied without reduction. If white paint only is desired, it should be purchased under TT-P-97.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Color required (see 3.3.2).
(c) Administrative provisions for inspection records (see 4.1).
(d) Level of packaging and packing required (see 5.1).
(e) Size of unit container (see 5.1).
(f) Types (see 1.2).

6.3 Basis of purchase. The paint should be purchased by volume, the unit being a gallon of 231 cubic inches at 15.6°C. (60°F.).

6.4 Suitable tinting pigments. Suitable tinting pigments include the following: brown iron oxide, red iron oxide, lamp black, phthalocyanine blue, phthalocyanine green, yellow iron oxide, chromium oxide, nickel titanate yellow, and carbon black. (These tinting systems should not be based on alkyd vehicles.)

TABLE III. Suggested type I formulations for two Tinted masonry paints

	lbs./100 gallons Blue	Pink
Titanium dioxide (rutile nonchalking)	141.5	116.5
Phthalo Blue	1.8	---
Calcium carbonate (low oil absorption)	236	261
Red iron oxide	---	1.7
Mica (Mineralite 3 - X)	68	71
Diatomaceous silica	63.5	71
Soya lecithin*	4	5
Thixcin R	4.6	4.7
Pliolite AC	94	94
Chlorinated paraffin 40 percent	47	47
Chlorinated paraffin 70 percent	47	47
Solvesso 100	203	199
Mineral spirits	203	199
Total	1113.4	1116.9
PVC	52%	53%
Solids (wt. percent)	63.5%	64.4%

* Variable dependent upon color system used

3 to 10 lbs. soya lecithin per 100 gals. is recommended range of use.

NOTE: Above formulation is not recommended for use as pastel yellow.
For pastel yellow use type II formulation as suggested in table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:18Z
Check the source to verify that this is the current version before use.

TABLE IV. Formulations for deep tone

Pigments	Type II**									
	Blue		Brown		Green		Red.		Yellow	
	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
Titanium dioxide (rutile)	25.2	28.5	--	--	20.5	22.5	--	--	--	--
Silica, diatomaceous	12.0	13.0	12.5	13.5	13.0	14.0	13.0	14.0	12.5	13.5
Calcium carbonate	44.0	49.0	50.0	55.0	46.5	51.5	47.5	52.5	44.5	49.5
Mica (Mineralite 3x)	12.5	13.5	12.5	13.5	14.0	15.0	13.0	14.0	13.0	14.0
Lamp black	*	*	--	--	*	*	--	--	--	--
Brown iron oxide	--	--	20.5	22.5	--	--	--	--	--	--
Red iron oxide	--	--	--	--	--	--	22.0	24.0	--	--
Phthalocyanine blue	*	*	--	--	--	--	--	--	--	--
Phthalocyanine green	--	--	--	--	*	*	--	--	--	--
Nickel titanate yellow	--	--	--	--	*	*	--	--	25.5	28.5

* Tinting pigments --As needed to give desired color.
** See table V for suggested formulae.

TABLE V

Suggested Type II Formulae

Deep-tone blue		Deep-tone brown		Sun-yellow	
Rutile TiO_2	153.0	Brown iron oxide	115	Nickel titanate yellow	142
Phthalo blue toner	3.0	Calcium carbonate	288	Calcium carbonate	250
Lampblack	0.4	Mica (Mineralit 3X)	72	Mica (Mineralite 3X)	70
Calcium carbonate	260.0	Dia. silica	72	Dia. silica	70
Mica (Mineralite 3X)	75.0	Thixcin R	5	Thixcin R	5
Diatomaceous silica (Celite 110)	70.0	Pliolite AC	98	Pliolite AC	94
Thixcin R	5.0	CP 40	49	CP 40	47
Pliolite AC	100.0	CP 70	49	CP 70	47
Chlorinated paraffin 40	50.0	Lecithin	10	Lecithin	2
Chlorinated paraffin 70	50.0	Hi flash naphtha	194	Hi flash naphtha	203
Soya lecithin	2.0	Mineral spirits	194	Mineral spirits	203
High flash naphtha	193.0				
Mineral spirits	193.0				
	1154.4		1146		1133

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:18Z
Check the source to verify that this is the current version before use.

TABLE V

Suggested Type II Formulae (con.)

Deep-tone green		Deep-tone red	
Rutile TiO_2	120	Red iron oxide	125
Phthalo green	10	Calcium carbonate	275
Lampblack	1	Mica (Mineralite 3x)	75
Calcium carbonate	275	Dia. silica	75
Mica (Mineralite 3x)	80	Thixcin R	5
Dia. silica	75	Pliolite AC	96
Thixcin	5	CP 40	48
Pliolite AC	100	CP 70	48
CP 40	50	Lecithin	10
CP 70	50	Hi flash naphtha	200
Lecithin	2	Mineral spirits	200
Hi flash naphtha	190		
Mineral spirits	190		
	1148		1157

TABLE VI. Fill coat formula

Charge into change can mixer:	
Pliolite S-5A	62
Chlorowax 40	33
Mineral spirits	65
Solvesso 100 (R)	65
Nuoperse 657	8
Mix until resin is completely dissolved, and then add slowly:	
TR asbestos shorts	50
Superwhite silica	400
Duramite	200
Bentone 38	5
Mineral spirits	134
Solvesso 100 (R)	134
Methanol (Methyl alcohol)	2

MILITARY CUSTODIANS

Navy-Yd
Air Force - 84

Review activities:

Navy-Yd
Air Force - 84, 85

User activities:

MI
HAA

CIVIL AGNECY INTERESTS:

GSA-FSS
HUD-HAA

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 10 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:18Z
Check the source to verify that this is the current version before use.